

Hanxiang Wang

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Education

University of Auckland – Engineering and Design
Bachelor's of Engineering (Software)

Expected May 2028
GPA: 8.75/9

Skills

Programming Languages: Python, Java, HTML, CSS, MATLAB, C, C#, C++

Technologies: Git, MATLAB, Microchip Studio, VS Code, MS Office, Google Suite, Google Cloud/AWS, FL Studio

Frameworks and Libraries: PyTorch, JavaFX, Pygame, TensorFlow, OpenAI tools, discord.py

Projects

DOTS AI (Genetic Algorithm Sandbox)

[Project GitHub Page](#)

- Utilises genetic algorithms in Python to simulate natural selection to improve a computer-simulated dots.
- Sandbox environment created in Pygame.
- Implements user-level customisation of obstacles, checkpoints and endpoints with keyboard controls.
- Allows fine setting adjustments to the simulation.

Quiz Platformer Game

[Project GitHub Page](#)

- Made fully in Python using Pygame, implementing self-coded collision and physics systems.
- Utilises Python's requests package to generate questions through API calls.
- Implements lots of UI features, such as buttons, textboxes and indicator bars etc.
- Implements external file reading to allow the user to customise keybinds and save settings.

Trial Simulation

[Project GitHub Page](#)

- Utilises JavaFX with Scenebuilder for UI design and interaction.
- Implements OpenAI chat completion with API calls to create interactable NPCs with multi-threading for continuity.
- Utilises JavaFX Keyframes to create timers and visual animation, such as fading and translation for UI smoothness.
- Uses OOP in Java to create a hierarchy of classes.

Reinforcement Learning Snake AI

[Project GitHub Page](#)

- Utilises Keras and OpenAI Gym to create a sandbox environment where an Agent plays Snake and improves.
- Visualisation of the sandbox environment created in Pygame.
- Implements a simple Neural Network to feed game state information to the Agent.
- Fine-tuning of rewards to optimise the outcome over 500,000 steps and 50 episodes.

Extracurricular

E-Sports Club Council Member, Botany Downs Secondary College.

Feb 2022 – Dec 2023

- Moderated a Discord server with over 150+ members.
- Hosted game night events for the server with more than 20 participants.
- Collaborated with other council members to host events or plan ideas.

Experience

Customer Service Representative, bp

Feb 2023 – Present

- Served thousands of customers over many years with exceptional customer service.
- Completed coffee orders while simultaneously managing checkout queue.
- Managed store inventory, checked expiration dates and restocked store.
- Communicated well and had great vibes with coworkers.